

Fuel Efficiency in Motor Vehicles: An Exploratory Report

A Sample Report Demonstrating Quarto Features

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Note

This is a sample, not an assignment. Comments that start with # explain what Quarto is doing. You can delete them when you write your own report. Render the file from an RStudio project. If PDF fails, change `format: pdf` to `format: html` and click Render again.

1 Introduction

Fuel efficiency is one of the most important factors car buyers consider. In this report, we explore the relationship between a vehicle's weight and its fuel efficiency using the classic `mtcars` dataset, which contains specifications for 32 car models from a 1974 *Motor Trend* magazine issue.

Notice that the number of cars above (32) was **not typed manually** — it was inserted with inline R code (``r nrow(mtcars)``). This is one of the most useful habits to build: numbers that come from your data should be computed, not hard-coded, so the report updates automatically if the data changes.

2 Data Overview

2.1 Variable Description

Table 1 describes the variables used in this report.

```
# label + tbl-cap together create a numbered table you can cite with @tbl-...

var_table <- tibble(
  Variable = c("mpg", "wt", "hp", "cyl"),
  Description = c(
    "Miles per gallon (fuel efficiency)",
    "Weight (in 1000 lbs)",
    "Gross horsepower",
    "Number of cylinders"
  )
)

kable(var_table) # knitr::kable() prints a clean table in HTML or PDF
```

Table 1: Description of key variables in the `mtcars` dataset

Variable	Description
mpg	Miles per gallon (fuel efficiency)
wt	Weight (in 1000 lbs)
hp	Gross horsepower
cyl	Number of cylinders

2.2 A Quick Look at the Data

Here are the first few rows of the raw data, shown in Table 2.

```
kable(head(mtcars[, c("model", "mpg", "wt", "hp", "cyl")))
```

Table 2: First six cars in the dataset

model	mpg	wt	hp	cyl
Mazda RX4	21.0	2.620	110	6
Mazda RX4 Wag	21.0	2.875	110	6
Datsun 710	22.8	2.320	93	4
Hornet 4 Drive	21.4	3.215	110	6
Hornet Sportabout	18.7	3.440	175	8
Valiant	18.1	3.460	105	6

3 Exploratory Analysis

3.1 Distribution of Fuel Efficiency

The average fuel efficiency across all cars is 20.1 miles per gallon, with a standard deviation of 6. The two numbers in that sentence are inline R, not typed by hand. Figure 1 shows the full distribution.

```
# label + fig-cap create a numbered figure you can cite with @fig-...

ggplot(mtcars, aes(x = mpg)) +
  geom_histogram(binwidth = 3, fill = "steelblue", color = "white") +
  labs(x = "Miles per gallon", y = "Count") +
  theme_minimal()
```

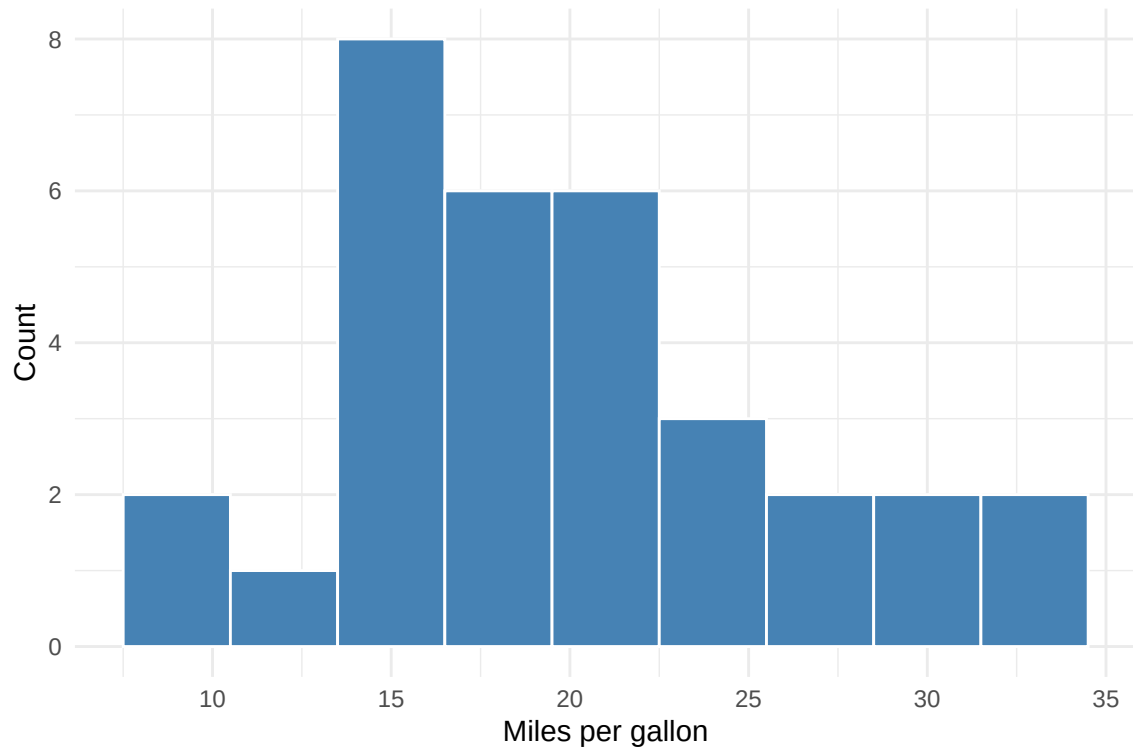


Figure 1: Distribution of miles per gallon across all cars

3.2 Weight, Horsepower, and Efficiency

To compare two relationships at once, we place two plots side by side using `layout-ncol`, shown in Figure 2.

```
# Two ggplot() calls in one chunk + layout-ncol: 2 → two panels.  
# fig-subcap labels each panel. The chunk still has one overall @fig-side.  
  
ggplot(mtcars, aes(x = wt, y = mpg)) +  
  geom_point(color = "darkred") +  
  labs(x = "Weight (1000 lbs)", y = "MPG") +  
  theme_minimal()  
  
ggplot(mtcars, aes(x = hp, y = mpg)) +  
  geom_point(color = "darkgreen") +  
  labs(x = "Horsepower", y = "MPG") +  
  theme_minimal()
```

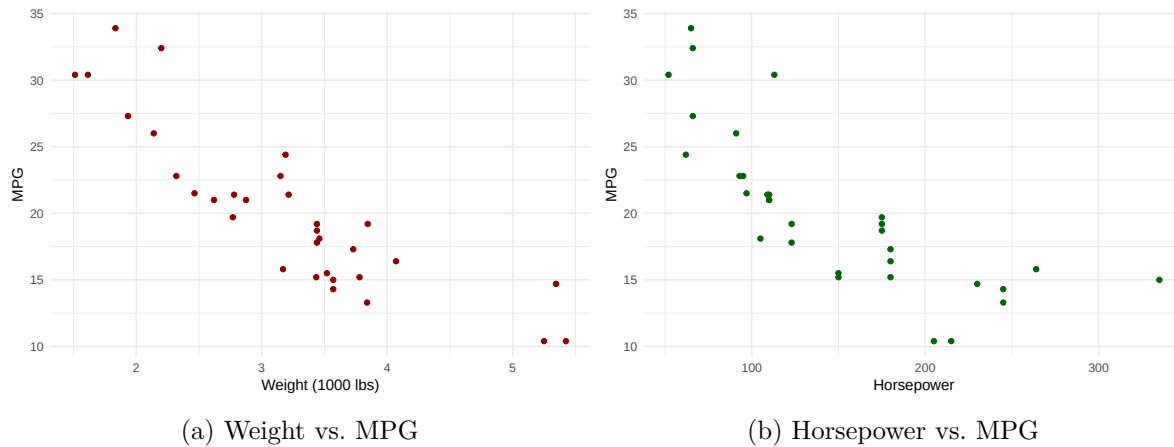


Figure 2: Weight and horsepower versus fuel efficiency

Both panels suggest a **negative relationship**: heavier, more powerful cars tend to be less fuel efficient.

4 Statistical Model

4.1 Model Specification

We fit a simple linear regression of fuel efficiency on weight:

$$\text{mpg}_i = \beta_0 + \beta_1 \cdot \text{wt}_i + \varepsilon_i$$

where β_0 is the intercept, β_1 is the slope, and ε_i is the error term for car i .

4.2 Model Results

```
model <- lm(mpg ~ wt, data = mtcars)
# broom::tidy() puts the coefficients in a data frame kable() can print.
kable(broom::tidy(model), digits = 3)
```

Table 3: Linear regression results: MPG explained by weight

term	estimate	std.error	statistic	p.value
(Intercept)	37.285	1.878	19.858	0
wt	-5.344	0.559	-9.559	0

The fitted model can be written as:

$$\widehat{\text{mpg}} = 37.29 - 5.34 \times \text{wt}$$

This means that for every additional 1,000 lbs of weight, fuel efficiency drops by about 5.34 miles per gallon, holding all else constant.

4.3 Model Fit

```
ggplot(mtcars, aes(x = wt, y = mpg)) +
  geom_point(color = "steelblue") +
  geom_smooth(method = "lm", se = TRUE, color = "black") +
  labs(x = "Weight (1000 lbs)", y = "Miles per gallon") +
  theme_minimal()
```

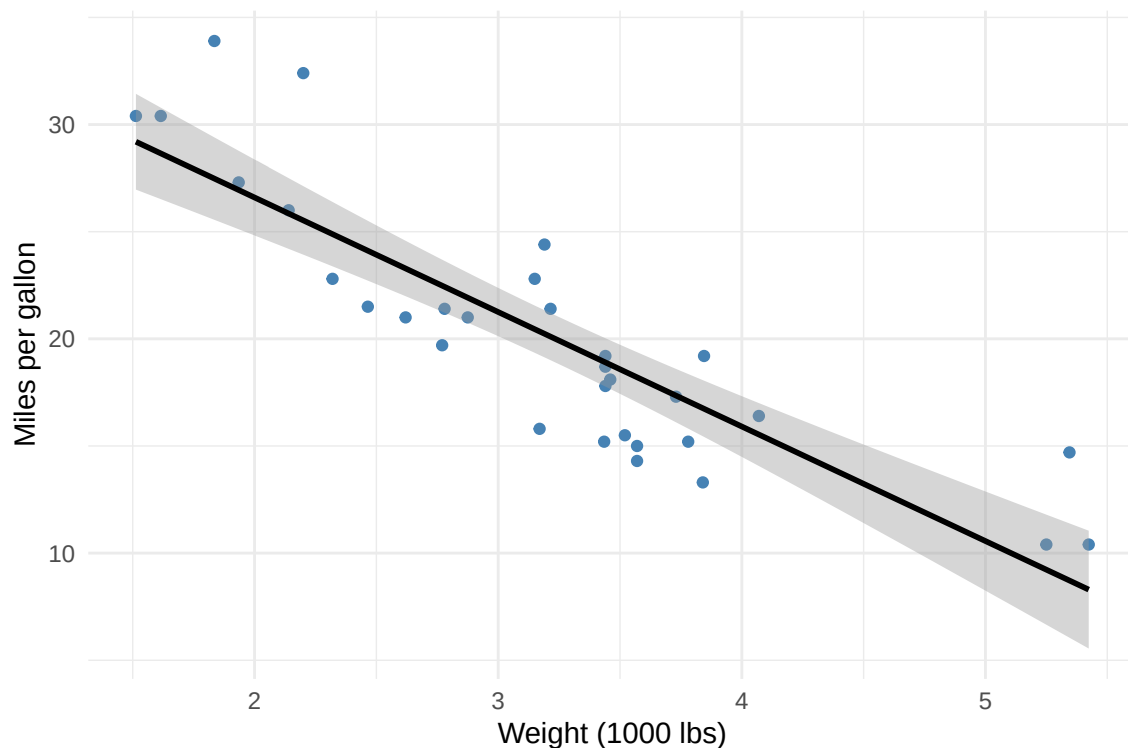


Figure 3: Fitted regression line over the observed data

The model explains $R^2 = 0.75$ of the variation in fuel efficiency — a strong relationship for a single-variable model.

5 Conclusion

This short analysis shows that vehicle weight is a strong predictor of fuel efficiency: heavier cars consistently get fewer miles per gallon. A natural next step would be to extend the model in Section [5.1](#).

5.1 Possible Extensions

Future work could include:

- Adding horsepower and cylinder count as additional predictors
- Comparing automatic vs. manual transmissions
- Checking whether the relationship is linear or curved at extreme weights

References

Henderson, H. V. and Velleman, P. F. (1981). *Building multiple regression models interactively*. Biometrics, 37, 391-411. (Source of the `mtcars` dataset, distributed with base R.)